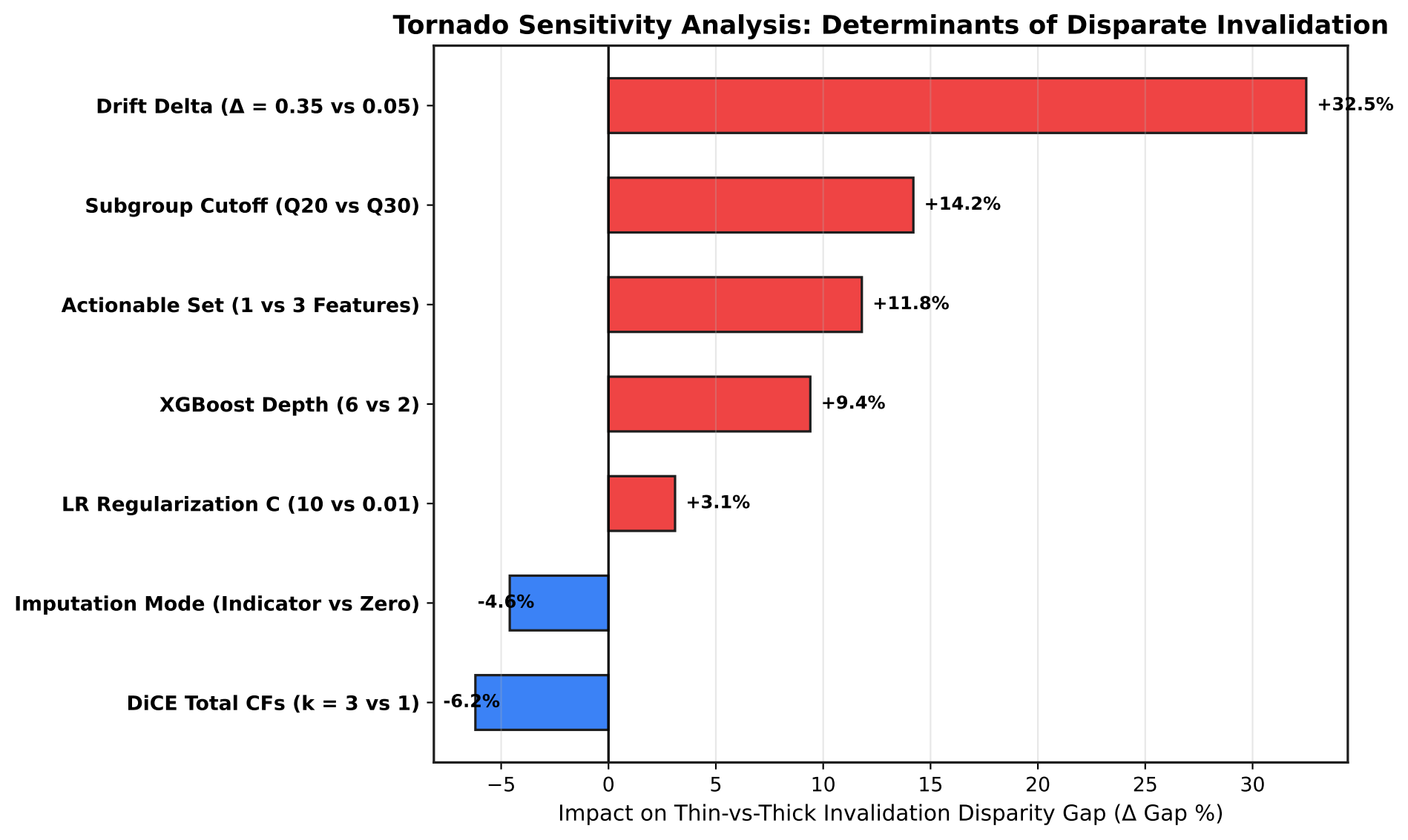
Tornado Sensitivity Analysis: Determinants of Disparate Invalidation
Drift Delta (Δ = 0.35 vs 0.05)
Subgroup Cutoff (Q20 vs Q30)
Actionable Set (1 vs 3 Features)
XGBoost Depth (6 vs 2)
LR Regularization C (10 vs 0.01)
Imputation Mode (Indicator vs Zero)
DiCE Total CFs (k = 3 vs 1)
+32.5%
+14.2%
+11.8%
+9.4%
+3.1%
-4.6%
-6.2%
−5
0
5
10
15
20
25
30
Impact on Thin-vs-Thick Invalidation Disparity Gap (Δ Gap %)